ModelIt!

Teacher's Guide

The Nitrogen Crisis: Why Fertilizer Is Destroying Our Rivers

Modeling Nutrient Pollution and Aquatic Ecosystem Collapse

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-LS2-4, HS-ESS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-4, HS-ESS2-2

Use mathematical representations to support claims for the cycling of matter and flow of energy among organisms in an ecosystem. Analyze geoscience data to make the claim that one change to Earth's surface can create feedbacks that cause changes to other Earth's systems.

Three-Dimensional Learning

Science Practice	**Using Mathematical Representations** — Students use mathematical relationships to model nitrogen flow from agricultural application through runoff, biological amplification, and oxygen depletion to predict dead zone formation.
Disciplinary Core Idea	**LS2.B Cycles of Matter and Energy Transfer / ESS2.A Earth Materials and Systems** — Matter cycles between the living and nonliving components of ecosystems; human activities can disrupt nutrient cycles with cascading effects on Earth's systems.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from individual farm fertilizer decisions through watershed-scale nutrient transport to continental-scale dead zone formation, identifying leverage points for intervention.

CAST Testing Alignment

MULTIPLE CHOICE:

A farmer applies 200 kg of nitrogen fertilizer per hectare. Based on your model, approximately how much of that nitrogen will eventually reach the nearest river as runoff?

CONSTRUCTED RESPONSE:

Using your model, explain the complete causal chain from a farmer applying fertilizer in Iowa to a dead zone forming in the Gulf of Mexico 1,500 miles downstream. Reference Nitrogen Runoff, Algal Bloom Growth, and Oxygen Depletion in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The invention of synthetic nitrogen fertilizer in 1913 (the Haber-Bosch process) is arguably the most impactful chemical innovation in human history — it enabled the feeding of billions of people who would otherwise have starved. But it also created a nitrogen pollution crisis that is destroying aquatic ecosystems worldwide. Understanding this trade-off is one of the defining challenges of the 21st century.

The Haber-Bosch Revolution

Before 1913, all nitrogen for agriculture came from natural sources — manure, crop rotation with legumes, and guano. The Haber-Bosch process converts atmospheric nitrogen gas (N2) into ammonia (NH3) using high temperature and pressure, making unlimited synthetic fertilizer possible. Today, approximately 50% of the nitrogen atoms in your body were fixed by the Haber-Bosch process. It has enabled the global population to grow from 1.7 billion to 8 billion. But it has also doubled the amount of reactive nitrogen cycling through Earth's ecosystems.

The Runoff Cascade

When nitrogen fertilizer is applied to fields, crops absorb only 30-50% under best management practices. The rest has three fates: it volatilizes into the atmosphere as ammonia or nitrous oxide (a potent greenhouse gas), it leaches into groundwater contaminating drinking water supplies, or it washes off the soil surface during rainfall as dissolved nitrate. This runoff enters streams, flows through river systems, and accumulates in lakes and coastal waters. The Mississippi River basin drains 41% of the continental United States — every farm in this vast watershed contributes to the nutrient load that eventually reaches the Gulf of Mexico.

Eutrophication and Dead Zones

When excess nitrogen reaches a water body, it fertilizes algae just as it fertilizes crops. Algal populations explode, doubling every 1-3 days and forming massive blooms that can cover hundreds of square kilometers. When these algae die, bacteria decompose the biomass and consume dissolved oxygen in the process. When oxygen drops below 2 mg/L (hypoxia), fish flee if they can and suffocate if they cannot. The resulting dead zone is a biological desert. The Gulf of Mexico dead zone averages 14,000 square kilometers annually — larger than Connecticut. There are now over 500 dead zones worldwide, up from 49 in the 1960s.

Solutions and Trade-offs

Solving the nitrogen crisis requires balancing food production with environmental protection. Precision agriculture uses GPS-guided equipment and soil sensors to apply exactly the nitrogen each section of field needs — reducing application by 15-30% without yield loss. Cover crops prevent bare soil from eroding and capture residual nitrogen between growing seasons. Riparian buffer strips (vegetated zones along waterways) intercept and filter runoff before it reaches streams. Constructed wetlands can denitrify water through bacterial processes. The technology exists — the challenge is implementation at scale across millions of farms.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of the nitrogen pollution system: Fertilizer Application, Nitrogen Runoff, Algal Bloom Growth, Oxygen Depletion, Fish Kill Rate, Soil Depletion, and Water Treatment Cost — with Fertilizer Application as the external driver.

E **Establish Relationships**

Students determine that Fertilizer Application drives Nitrogen Runoff (20-40% loss), which triggers Algal Bloom Growth. Dead algae cause Oxygen Depletion, leading to Fish Kill Rate. Soil Depletion increases fertilizer dependence. Water Treatment Cost rises with pollution severity.

V **Visualize & Model**

Students build a computational model showing the cascade from farm field to dead zone and the economic feedback through Water Treatment Cost.

E **Evaluate Outcomes**

Students run conventional farming, precision agriculture, and soil regeneration scenarios to compare environmental and economic outcomes and identify the most effective intervention points.

R **Revise & Extend**

Students add Wetland Filtration, Atmospheric Deposition, or Phosphorus Loading to explore how natural buffers and additional nutrient sources affect dead zone dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"That bright green water isn't paint. It's trillions of algae, fed by fertilizer from farms a thousand miles away. And underneath that green surface, everything is dying."*

Watch For

Show satellite images of the Gulf of Mexico dead zone and compare to a map of the Mississippi River watershed. Ask: How does fertilizer from Iowa end up killing fish in Louisiana?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling how a farmer's perfectly reasonable decision to fertilize crops creates an environmental disaster 1,500 miles downstream."*

Watch For

Have students read objectives. Pre-teach 'eutrophication' and 'hypoxia.' Quick-write: What do you think happens to fertilizer when it rains?

Slide 3: Big Question

3 min

Say: *"Fertilizer helps plants grow. More plants, more food, more life. So why does fertilizer in water kill everything?"*

Watch For

Students brainstorm: When is 'more nutrients' actually harmful? Can you think of other examples where too much of a good thing becomes destructive?

Slide 4: LEVER Framework

2 min

Say: *"We're tracing a nitrogen atom from a fertilizer bag in Iowa to a dead fish in the Gulf of Mexico. The journey will change how you think about agriculture."*

Watch For

Preview each LEVER step. Emphasize that this model connects individual farm decisions to continental-scale environmental consequences.

Slide 5: Activity 1: Components

8 min

Say: *"One decision: how much fertilizer to apply. The consequences cascade through seven components across 1,500 miles."*

Watch For

Guide component sorting. Discuss why Fertilizer Application is external (farmer's decision) while all downstream effects are internal ecological and economic responses.

Slide 6: Activity 2: Connections

8 min

Say: *"Follow the nitrogen. Field to stream. Stream to river. River to gulf. Algae to dead zone. Dead zone to fish kills. Fish kills to economic collapse. It's all connected."*

Watch For

Help students trace the complete cascade. Identify the Soil Depletion vicious cycle. Calculate how much nitrogen reaches the ocean from a single farm.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three approaches to farming, three very different outcomes for the Gulf of Mexico. Let's find out which one actually works."*

Watch For

Students predict dead zone size for each scenario. Run simulations. Compare environmental AND economic outcomes — including Water Treatment Cost.

Slide 8: Discoveries

5 min

Say: *"The Haber-Bosch process feeds half the world. It also created 500 dead zones. Can we keep the food and fix the water?"*

Watch For

Lead discussion on the food-environment trade-off. Connect to the real Gulf of Mexico dead zone data. Discuss solutions that are available but not widely adopted.

Slide 9: STEM Challenge

12 min

Say: *"The EPA says your community must cut nitrogen runoff by 40% in five years or face mandatory restrictions. Design the plan that saves both the farms and the fish."*

Watch For

Groups design comprehensive watershed management plans. Must balance agricultural productivity with environmental targets. Present as proposals to a simulated EPA panel.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Fertilizer Application
- **INTERNAL:** Nitrogen Runoff, Algal Bloom Growth, Oxygen Depletion, Fish Kill Rate, Soil Depletion, Water Treatment Cost

Relationships

- **Fertilizer Application to Nitrogen Runoff:** POSITIVE (+) — Higher Fertilizer Application increases the amount of excess nitrogen available on field surfaces. During rainfall events, a greater fraction of this nitrogen dissolves and washes into waterways as nitrate runoff — typically 20-40% of applied nitrogen is lost.
- **Nitrogen Runoff to Algal Bloom Growth:** POSITIVE (+) — Higher Nitrogen Runoff delivers more dissolved nutrients to lakes and coastal waters. Since nitrogen is often the limiting nutrient in marine systems, increased supply removes the growth constraint on algae, triggering explosive population blooms.
- **Algal Bloom Growth to Oxygen Depletion:** POSITIVE (+) — Larger Algal Bloom Growth produces more biomass that eventually dies and sinks. Bacterial decomposition of this massive organic load consumes dissolved oxygen at rates that exceed atmospheric replenishment, driving dissolved oxygen to hypoxic levels.

Key Discoveries

- Only 30-50% of applied nitrogen fertilizer is actually absorbed by crops — the rest enters waterways as Nitrogen Runoff, making agriculture the largest source of nutrient pollution globally
- There is a threshold effect: moderate nitrogen loading allows rivers and oceans to process the excess, but above a critical concentration, Algal Bloom Growth overwhelms the system and dead zones form
- Soil Depletion creates a vicious cycle — degraded soil holds less nitrogen, requiring more fertilizer, which degrades soil further while increasing runoff
- The economic costs of nutrient pollution (water treatment, fishery losses, tourism damage) often exceed the economic gains from the excess fertilizer that caused the pollution

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fertilizer is artificial and therefore harmful"	Synthetic nitrogen fertilizer is chemically identical to nitrogen from natural sources — plants cannot distinguish between a nitrate ion from a compost pile and one from a fertilizer bag. The problem is not the chemistry but the quantity. Natural nitrogen cycling adds about 140 million tons of reactive nitrogen to ecosystems annually; the Haber-Bosch process adds another 120 million tons. We have doubled the nitrogen cycle, overwhelming ecosystems' capacity to process it.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Dead zones are permanent"	Most dead zones are seasonal — they form in late summer when warm water holds less oxygen and algal die-offs are largest, then dissipate in fall and winter when cooler temperatures and storms mix oxygen back into the water. This means the damage is potentially reversible if nutrient loading is reduced. The Black Sea dead zone — once the largest in the world — largely disappeared after the collapse of Soviet agriculture reduced fertilizer use. Recovery is possible, but it requires sustained reduction in nutrient inputs.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Organic farming solves the nitrogen problem"	Organic farming reduces synthetic fertilizer use but does not eliminate nitrogen runoff. Organic nitrogen sources (manure, compost, cover crop residues) can still wash into waterways if applied in excess or during rain events. In some cases, manure over-application on organic farms creates worse runoff than well-managed conventional fields. The solution is not the nitrogen source but the nitrogen balance — applying only what crops can absorb, regardless of whether the source is synthetic or organic.